

Unit 24: Area and Estimating with Finite Sums — Full Solutions

Warm-up

1. $f(x) = x^2$ on $[0, 4]$, $n = 4$, left sum.

$\Delta x = \frac{4-0}{4} = 1$. Left endpoints: $x = 0, 1, 2, 3$. Function values: $f(0) = 0$, $f(1) = 1$, $f(2) = 4$, $f(3) = 9$.

$$L_4 = 1 \times (0 + 1 + 4 + 9) = 14$$

$$L_4 = 14.$$

2. Same setup, right sum.

Right endpoints: $x = 1, 2, 3, 4$. Function values: $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, $f(4) = 16$.

$$R_4 = 1 \times (1 + 4 + 9 + 16) = 30$$

$$R_4 = 30.$$

3. $f(x) = x + 1$ on $[0, 4]$, $n = 4$, left sum.

$\Delta x = 1$. Left endpoints: $0, 1, 2, 3$. Values: $1, 2, 3, 4$.

$$L_4 = 1 \times (1 + 2 + 3 + 4) = 10$$

$$L_4 = 10.$$

4. Same setup, right sum.

Right endpoints: $1, 2, 3, 4$. Values: $2, 3, 4, 5$.

$$R_4 = 1 \times (2 + 3 + 4 + 5) = 14$$

$$R_4 = 14.$$

5. $f(x) = x^2$ on $[0, 2]$, $n = 4$, left sum.

$\Delta x = \frac{2-0}{4} = 0.5$. Left endpoints: $0, 0.5, 1, 1.5$. Values: $0, 0.25, 1, 2.25$.

$$L_4 = 0.5 \times (0 + 0.25 + 1 + 2.25) = 0.5 \times 3.5 = 1.75$$

$$L_4 = 1.75.$$

6. Same setup, right sum.

Right endpoints: $0.5, 1, 1.5, 2$. Values: $0.25, 1, 2.25, 4$.

$$R_4 = 0.5 \times (0.25 + 1 + 2.25 + 4) = 0.5 \times 7.5 = 3.75$$

$$R_4 = 3.75.$$

Standard

7. $f(x) = x^2$ on $[0, 4]$, $n = 4$, midpoint sum.

$\Delta x = 1$. Midpoints: 0.5, 1.5, 2.5, 3.5. Values: 0.25, 2.25, 6.25, 12.25.

$$M_4 = 1 \times (0.25 + 2.25 + 6.25 + 12.25) = 21$$

$$M_4 = 21.$$

8. $f(x) = 9 - x^2$ on $[0, 3]$, $n = 3$.

$\Delta x = 1$. Note f is decreasing on $[0, 3]$ (its derivative $-2x$ is negative for $x > 0$).

Left sum: endpoints 0, 1, 2; values 9, 8, 5.

$$L_3 = 1 \times (9 + 8 + 5) = 22$$

Right sum: endpoints 1, 2, 3; values 8, 5, 0.

$$R_3 = 1 \times (8 + 5 + 0) = 13$$

$L_3 = 22$, $R_3 = 13$. Since f is decreasing, the left sum overestimates and the right sum underestimates the true area — consistent with our rule.

9. $f(x) = \frac{1}{x}$ on $[1, 5]$, $n = 4$, left sum.

$\Delta x = 1$. Left endpoints: 1, 2, 3, 4. Values: 1, 0.5, 0.3333, 0.25.

$$L_4 = 1 \times (1 + 0.5 + 0.3333 + 0.25) \approx 2.0833$$

$$L_4 \approx 2.0833.$$

10. $f(x) = \sqrt{x}$ on $[0, 4]$, $n = 4$, right sum.

$\Delta x = 1$. Right endpoints: 1, 2, 3, 4. Values: 1, 1.4142, 1.7321, 2.

$$R_4 = 1 \times (1 + 1.4142 + 1.7321 + 2) \approx 6.1463$$

$$R_4 \approx 6.1463.$$

11. $f(x) = x^2 + 1$ on $[-1, 1]$, $n = 4$, midpoint sum.

$\Delta x = \frac{1 - (-1)}{4} = 0.5$. Subintervals: $[-1, -0.5]$, $[-0.5, 0]$, $[0, 0.5]$, $[0.5, 1]$. Midpoints: $-0.75, -0.25, 0.25, 0.75$.

Values: $f(-0.75) = 0.5625 + 1 = 1.5625$; $f(-0.25) = 0.0625 + 1 = 1.0625$; $f(0.25) = 1.0625$; $f(0.75) = 1.5625$.

$$M_4 = 0.5 \times (1.5625 + 1.0625 + 1.0625 + 1.5625) = 0.5 \times 5.25 = 2.625$$

$$M_4 = 2.625.$$

12. $f(x) = 4 - x$ on $[0, 4]$, $n = 4$.

$\Delta x = 1$. Left sum: endpoints 0, 1, 2, 3; values 4, 3, 2, 1.

$$L_4 = 1 \times (4 + 3 + 2 + 1) = 10$$

Right sum: endpoints 1, 2, 3, 4; values 3, 2, 1, 0.

$$R_4 = 1 \times (3 + 2 + 1 + 0) = 6$$

Exact area: this region is a triangle with base 4 and height 4: $\text{Area} = \frac{1}{2}(4)(4) = 8$.

$L_4 = 10$ (overestimate, since f is decreasing), $R_4 = 6$ (underestimate), and the true area 8 falls right between them — exactly as expected.

Challenge

13. $f(x) = x^2$ on $[1, 3]$, $n = 5$, left sum.

$\Delta x = \frac{3-1}{5} = 0.4$. Left endpoints: 1, 1.4, 1.8, 2.2, 2.6. Values: 1, 1.96, 3.24, 4.84, 6.76.

$$L_5 = 0.4 \times (1 + 1.96 + 3.24 + 4.84 + 6.76) = 0.4 \times 17.8 = 7.12$$

$$L_5 = 7.12.$$

14. $f(x) = x^3$ on $[0, 2]$, $n = 4$, right sum.

$\Delta x = 0.5$. Right endpoints: 0.5, 1, 1.5, 2. Values: 0.125, 1, 3.375, 8.

$$R_4 = 0.5 \times (0.125 + 1 + 3.375 + 8) = 0.5 \times 12.5 = 6.25$$

$$R_4 = 6.25.$$

15. Why does the left sum underestimate and the right sum overestimate for an increasing function?

For an increasing function, on any subinterval, the function's smallest value occurs at the left edge of that piece (since the function is still climbing from there), and its largest value occurs at the right edge.

A left-endpoint rectangle uses that smallest height as its constant rectangle height. Since the actual curve is always at least as tall as that smallest value across the whole piece, the rectangle sits entirely underneath the curve — so the total left sum can never exceed the true area. This makes the left sum an underestimate.

A right-endpoint rectangle uses the largest height on that piece. Since the actual curve is never taller than that value anywhere in the piece, the rectangle sits entirely at or above the curve — so the total right sum can never fall short of the true area. This makes the right sum an overestimate.

16. $f(x) = \sin x$ on $[0, \pi]$, $n = 4$, midpoint sum.

$\Delta x = \frac{\pi}{4}$. Subintervals: $[0, \frac{\pi}{4}]$, $[\frac{\pi}{4}, \frac{\pi}{2}]$, $[\frac{\pi}{2}, \frac{3\pi}{4}]$, $[\frac{3\pi}{4}, \pi]$. Midpoints: $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$.

Values (using known sine values, in degrees for intuition: $22.5^\circ, 67.5^\circ, 112.5^\circ, 157.5^\circ$):

$$\sin\left(\frac{\pi}{8}\right) \approx 0.38268 \quad \sin\left(\frac{3\pi}{8}\right) \approx 0.92388$$

$$\sin\left(\frac{5\pi}{8}\right) \approx 0.92388 \quad \sin\left(\frac{7\pi}{8}\right) \approx 0.38268$$

(The middle two match the outer two due to the symmetry of sine around $\frac{\pi}{2}$.)

$$\text{Sum} = 0.38268 + 0.92388 + 0.92388 + 0.38268 = 2.61312$$

$$M_4 = \frac{\pi}{4} \times 2.61312 \approx 0.7854 \times 2.61312 \approx 2.0523$$

$M_4 \approx 2.0523$ (the true value of this area is exactly 2 — the midpoint rule gets remarkably close with just 4 rectangles).

17. Left sum = 20, right sum = 28, increasing function.

Since the function is increasing, the left sum underestimates and the right sum overestimates the true area, so:

$$20 \leq \text{true area} \leq 28$$

A better single estimate: average the two sums:

$$\frac{20 + 28}{2} = 24$$

Answer: the true area is somewhere between 20 and 28, and a better estimate using the average of the two sums is 24.

Applied

18. Velocity data, left sum, $\Delta t = 1$.

Left endpoints use $v(0), v(1), v(2), v(3)$: 0, 10, 18, 24.

$$L_4 = 1 \times (0 + 10 + 18 + 24) = 52$$

Answer: estimated distance = 52 feet.

19. Same data, right sum.

Right endpoints use $v(1), v(2), v(3), v(4)$: 10, 18, 24, 28.

$$R_4 = 1 \times (10 + 18 + 24 + 28) = 80$$

Answer: estimated distance = 80 feet. Since velocity is increasing throughout, the left sum (52 ft) underestimates and the right sum (80 ft) overestimates — the true distance traveled lies somewhere between the two.

20. Flow rate data, left sum, $\Delta t = 2$.

Left endpoints use $r(0), r(2), r(4), r(6)$: 5, 8, 12, 15.

$$L_4 = 2 \times (5 + 8 + 12 + 15) = 2 \times 40 = 80$$

Answer: estimated total water collected = 80 gallons.

21. Marginal profit data, right sum.

Right endpoints (for $n = 3$ subintervals over $[0, 3]$, with $\Delta t = 1$) use $P'(1), P'(2), P'(3)$: 120, 150, 180.

$$R_3 = 1 \times (120 + 150 + 180) = 450$$

Answer: estimated total profit accumulated = \$450.